

Seasonal Surge and Off-Season Slump: Analyzing the Variance in Demand for the Toronto Island Ferry Over 2025-2026*

Month by month comparison of the ticket sales and redemptions for the Toronto
Island ferry

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By analyzing the ticket sales and redemptions for the Toronto Island ferry provided by the City of Toronto, this study aims to identify how demand for the ferry changes throughout the year. The analysis finds that the ferry is far more popular during the summer months, demonstrating that there are peak months where sales are significantly higher than the baseline. This allows for the ferry services to optimize their staffing and vehicle scheduling over the course of a year.

1 Introduction

The Toronto Islands are the premier way to view the city's skyline. Thus, they become a hub for travelers to visit, whether they are just visiting Toronto or are from the surrounding area. Understanding when these visits happen, and what can influence the demand allows for the City of Toronto, which operates the ferries, to best optimize their spending versus returns.

The City of Toronto keeps records of how many ferry tickets are sold, and redeemed, in real time with servers being updated every 15 minutes. By taking these data points and aggregating them by what month and year the sales occurred, it was seen that the ferry is popular during the months of July and August, with November, December, January, February, and March being the least popular.

This trend of having busy summers and off-peak winters can be useful to the city as it allows their infrastructure to be equipped for their peak seasons by taking advantage of the off-peak to reduce the operating budget and schedule maintenance while minimizing lost

*Code and data are available at: https://github.com/KyleDong03/Toronto_Ferry_Sales.

revenue. Understanding the traffic patterns allows the City of Toronto to better decide how they can optimize their operations to save on costs, allowing them to reinvest more into their infrastructure and personnel.

Telegraphing paragraph: The remainder of this paper is structured as follows. Section 2 provides a description of the data used in the analysis.

2 Data

2.1 Data source

The Toronto Island ferry ticket sales and redemption data (2025-2026) (Fleet Services 2024) was collected by the fleet services department of the city of Toronto, published through Toronto Open Data (Gelfand 2025). This dataset was chosen as the Toronto Islands are increasing in popularity. With this data being the most up to date on the ticket sales and redemptions, which are the main interest when observing the Toronto Islands demand, this dataset became the optimal choice. Currently, the 32,000 most recent entries of the 279,573 total data points, to give a more focused study into the monthly, and subsequently seasonal trends.

The statistical programming language R (R Core Team 2023) was used to simulate the data, retrieve and clean the actual data, as well as test both the simulated and actual data. Tidyverse (Wickham et al. 2019), specifically lubridate (Grolemund and Wickham 2011), dplyr (Wickham et al. 2026), and testthat (Wickham 2011) were used throughout the process, with lubridate being used in the data cleaning to work with the date formatted data, dplyr being used to manipulate the data in the retrieval and cleaning, and testthat being utilized to form unit tests for the actual data. Opendatatoronto (Gelfand 2025) was used to retrieve the actual data, with a script that was also provided (Fleet Services 2024). Tinytable (Arel-Bundock 2026) and ggplot2 (Wickham 2016) were used to create the tables and graphs in this report, with scales (Wickham et al. 2025) being used to format the graphs.

2.2 Measurement

The 3 variables that were used in this analysis were the date, ticket sales, and ticket redemptions. Table 1 gives a glimpse into the data where every 15 minutes all the ticket information collected would be polled together, as seen in the Timestamp column which records the date and time of interest, and recorded on the system as totals. The tickets sold would be tracked by the online services or the ticket kiosks, and the tickets redeemed would be scanned when the ticket was scanned to be redeemed on site, where the totals would be recorded into the Redemption.Count and Sales.Count columns respectively.

Table 1: Toronto Island ferry ticket original data as retrieved from Open Data Toronto

Timestamp	Redemption.Count	Sales.Count
2026-08-22T10:45:00	9	2
2026-08-22T10:30:00	331	274
2026-08-22T10:15:00	277	199
2026-08-22T10:00:00	248	154
2026-08-22T09:45:00	200	181

The data was grouped by the month and year in which the record took place, and the ticket sales and redemptions were totalled to give the totals for the months between June 2025 and August 2026, thus giving 15 months to analyze, which can be seen in Table 2 and the respective columns. Thus the 3 variables, Data, Redemption_Count, and Sales_Count were constructed. Note that because the data chosen was between June 20, 2025 and August 22, 2026 each boundary month has partial month figures.

Table 2: Toronto Island ferry ticket data totaled by month and year

Date	Redemption Count	Sales Count
2025-06	102,757	104,945
2025-07	365,303	370,626
2025-08	380,782	385,275
2025-09	166,671	162,632
2025-10	85,523	81,714

2.3 Ticket Data

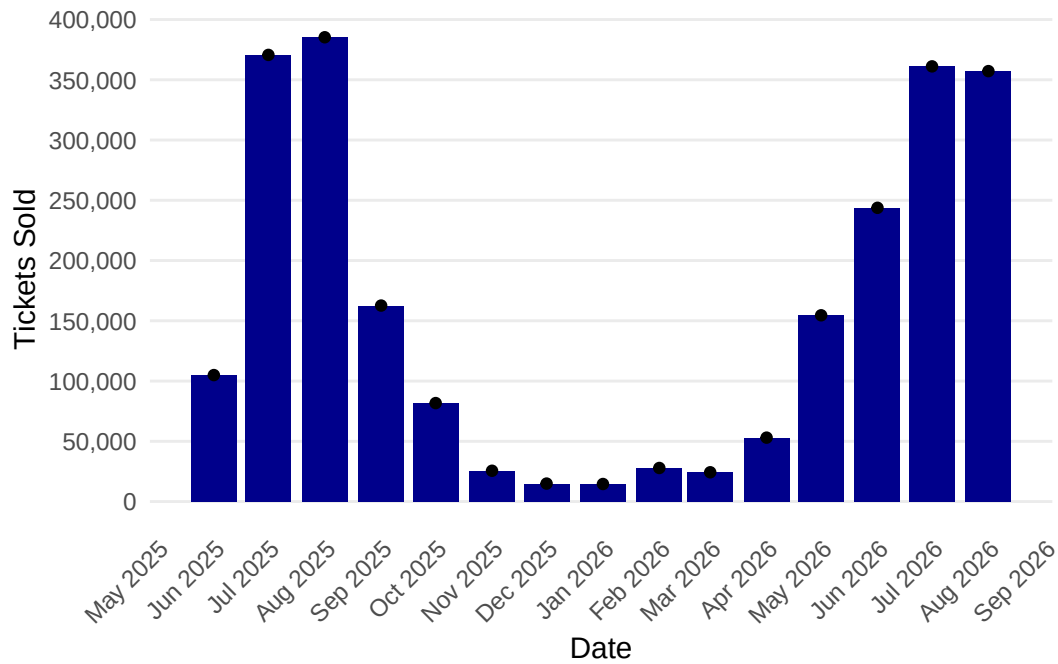


Figure 1: Total tickets sold during each month from June 2025 to August 2026

Figure 1 shows the amount of tickets sold every month through the specified dates giving an idea of what the seasonal trends are. It can be seen that the months of July and August are far higher than any other months, having a peak of 385,275 tickets sold in August 2025, with the least popular months being December and January, having a minimum of 14,572 tickets sold in January 2026.

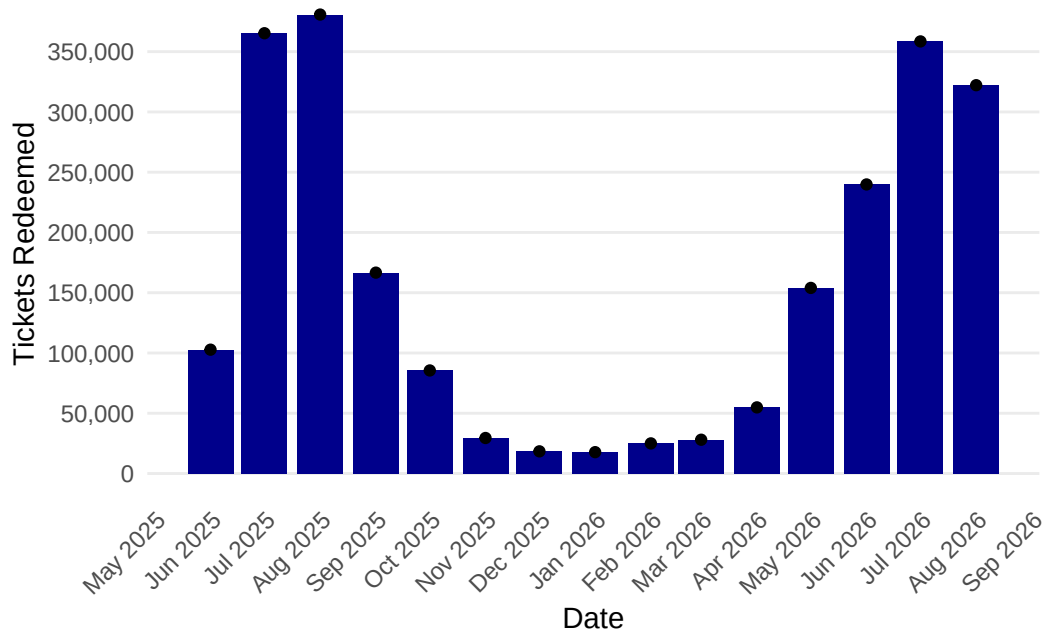


Figure 2: Total tickets redeemed during each month from June 2025 to August 2026

Figure 2 shows the amount of tickets redeemed every month through the specified dates giving an idea of what the seasonal trends are. It can be seen that the months of July and August are far higher than any other months, having a peak of 380,782 tickets sold in August 2025, with the least popular months being December and January, having a minimum of 17,745 tickets sold in January 2026. These trends and specific peaks match with the same months as the ticket sales, which makes sense as both are directly linked to each other, as tickets sold must be redeemed, and redeemed tickets must have been sold at some point.

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